Yuanshan Zhong

University at Buffalo | Biomedical Engineering

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Research Interests
Multimodal human physiological signal analysis and AI-driven neuroengineering, with interests in neural representation learning, brain–computer interfaces, neurostimulation, and closed-loop rehabilitation systems.

EDUCATION

University at Buffalo, SUNY *Aug 2025 – Present*

M.S. in Biomedical Engineering

University of Science and Technology of China (USTC) *Sep 2018 – Jun 2023*

B.S. in Computer Science

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RESEARCH EXPERIENCE

Research Assistant
Wenyao Xu Lab, University at Buffalo

Multimodal Neural Signal Analysis for Objective Biomarker Discovery in Auditory Disorders

University at Buffalo | Advisor: Prof. Wenyao Xu | Oct 2025 – Present

• Contributed to a multimodal neurophysiological system (SYNAPSE) integrating ear-EEG and pupillometry for objective assessment of sound sensitivity disorders (tinnitus, hyperacusis, misophonia).

• Utilized EEG preprocessing and LSL-based synchronization pipelines to enable temporally aligned multimodal analysis between neural and physiological signals.

• Extracted neural features (ERP, spectral power, cross-channel synchronization) and analyzed their relationships with physiological arousal signals and clinical questionnaire data.

• Built neural representation learning pipelines (EEGNex, Labram) and performed embedding analysis (PCA/UMAP), revealing interpretable latent structures linked to disease-related patterns and trial variability.

• Evaluated cross-subject generalization using leave-one-subject-out classification (~0.81 AUC) with label-shuffle validation, demonstrating robustness of learned representations.

Hybrid Learning for Robust Neural Control in Brain–Computer Interfaces

Independent Research | Apr 2025 – Present

• Designed an EEG-driven neural control framework for 2D cursor navigation using the BCI Competition IV-2a dataset, formulating a closed-loop control problem from neural signals.

• Developed a hybrid learning approach combining supervised neural decoding and reinforcement learning–based sequential decision-making to model the interaction between neural representations and control policies.

• Proposed a compact neural representation that compresses high-dimensional EEG features into a structured low-dimensional control state, integrating neural features, decoder outputs, and task geometry.

• Conducted cross-subject generalization experiments with strict ID/OOD splits, demonstrating improved training stability, convergence, and cross-subject generalization.

• Showed that compact neural representations improve robustness to EEG noise and enable stable and reliable neural control behavior.

• Investigating the robustness and cross-dataset generalizability of closed-loop neural control systems across multiple BCI datasets.

Modeling tissue heating in high-density microelectrode arrays for neurostimulation

University at Buffalo | Advisor: Prof. Filip Stefanovic | Aug 2025 – Present

• Developed finite-element models in COMSOL to simulate electric field and thermal effects in multilayer biological tissue for neurostimulation.

• Analyzed the effects of novel micro electrode arrays and stimulation parameters (current amplitude, pulse width, frequency) on tissue heating.

• Assessed electrode performance across various geometric configurations to identify optimized electrode sizes.

• Demonstrated that grid electrode designs maintain safety profiles comparable to conventional single electrodes.

Adaptive spatial optimization for high-density electrode grids that change size and shape

University at Buffalo | Advisor: Prof. Filip Stefanovic | Dec 2025 – Present

• Investigated high-density electrode arrays for selective stimulation of small muscles involved in dexterous movements.

• Modeled current distribution across different spatial electrode configurations for neurostimulation.

• Generated large-scale simulation data for electrode configuration analysis.

• Applied machine learning models (e.g., random forest and neural networks) to identify optimal electrode subsets for targeted stimulation.

Event–Frame Hybrid Neural Network for Occlusion Removal

University of Science and Technology of China | Advisor: Prof. Xueyang Fu | Dec 2022 – Jul 2023

• Developed a multimodal framework integrating event streams and image frames.

• Designed a hybrid neural network combining spiking neural networks (SNN) and convolutional networks.

• Developed cross-modal feature fusion to improve reconstruction under occlusion conditions.

• Constructed a dataset with 900+ samples and evaluated reconstruction performance using PSNR and SSIM.

Knowledge-Graph-Based Recommendation with Graph Representation Learning

Research Project | University of Science and Technology of China | 2022

• Studied the integration of knowledge graphs and collaborative filtering for explainable movie recommendation.

• Built a knowledge graph from Freebase entities linked to Douban movies and extracted multi-hop relational structures for representation learning.

• Implemented knowledge graph embedding models (TransE, TransR) and integrated them with matrix factorization to enhance item embeddings.

• Evaluated recommendation performance using ranking metrics (Recall@5/10, NDCG@5/10).

Programming Language Extension and Compiler Implementation

Research Project | University of Science and Technology of China | 2021

• Implemented a YACC-based parser to resolve ambiguity in relational expression grammars and enable correct evaluation of chained comparison expressions.

• Extended the educational PL/0 compiler with multi-dimensional array support, including symbol-table representation and runtime memory layout computation.

• Developed compiler backend mechanisms for array indexing, assignment, and control flow (for-loops and logical operators) through stack-machine instruction generation.

RESEARCH MANUSCRIPTS IN PREPARATION

Zhong, Y., Xu, W., et al. (2026).
Around-ear EEG analysis for stable neural feature discovery in auditory disorders.
Manuscript in preparation.

Zhong, Y., Stefanovic, F. (2026).
Finite-element modeling of electrical stimulation safety with grid electrodes.
Manuscript in preparation.

Zhong, Y., Stefanovic, F. (2026).
Optimization of high-density grid electrode stimulation for selective muscle activation.
Manuscript in preparation.

Zhong, Y. (2026).
Hybrid supervised–reinforcement learning for neural control systems.
Manuscript in preparation.

TECHNICAL SKILLS

Neuroengineering
EEG signal processing, neural representation learning, ERP analysis, spectral analysis

Machine Learning
PyTorch, deep learning, self-supervised learning, reinforcement learning

Modeling & Simulation
COMSOL Multiphysics, finite-element modeling

Programming
Python, MATLAB

Data Analysis
NumPy, SciPy, scikit-learn, MNE-Python

HONORS & AWARDS

National Scholarship (Third Prize), University of Science and Technology of China, 2018
Third Prize, Provincial Level Physics High School Competition (China)
First Prize, City-Level Mathematics High School Competition (China)